

東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

Hibernation Challenges and Optimization

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Background

- Hibernation is typically disabled by default in major distros
- Hibernation tests are requirements for customers
- 1500+ cycles on a fleet of systems
  - Takes several days to complete
  - Always challenging
  - Fails after several days on random systems
  - Random lockups

- Uncovers issues in firmware, core and drivers

Examples: Recent Failures

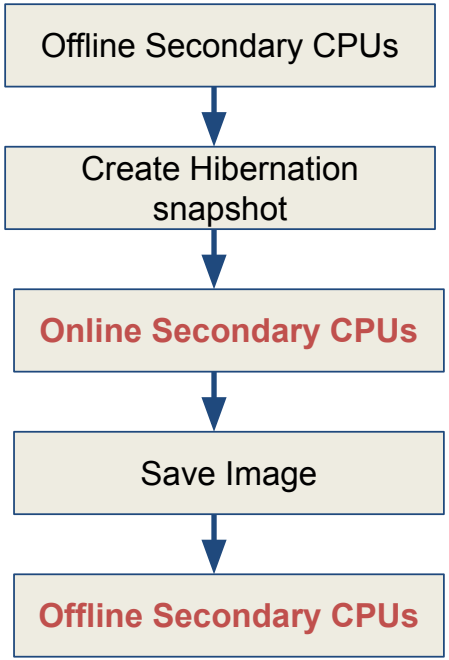
- PCI PM non atomic BAR address restore after D3
  - Memory Space Enable (MSE) bit is set
- printk calls after APIC disable cause lockup
  - klogd workers being invoked

Objective

- Internal changes
  - Reduce the cycle time
  - Increase fault tolerance
  - Add debug capabilities
- Are they useful for upstream?

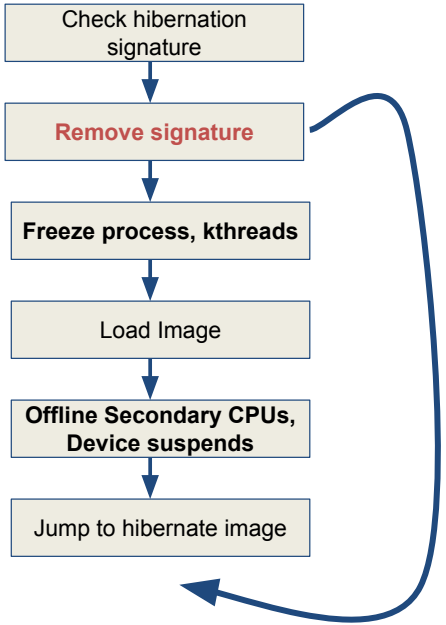
Reduce suspend time

- Wake enough CPUs to compress and write image (user configurable)
- Reduces suspend time by more than 60% on a workstation



Late discard of hibernation image

- Many fault paths after hibernation image is discarded
- Allow to retry image
- Optional (kernel command line)



Backup Image

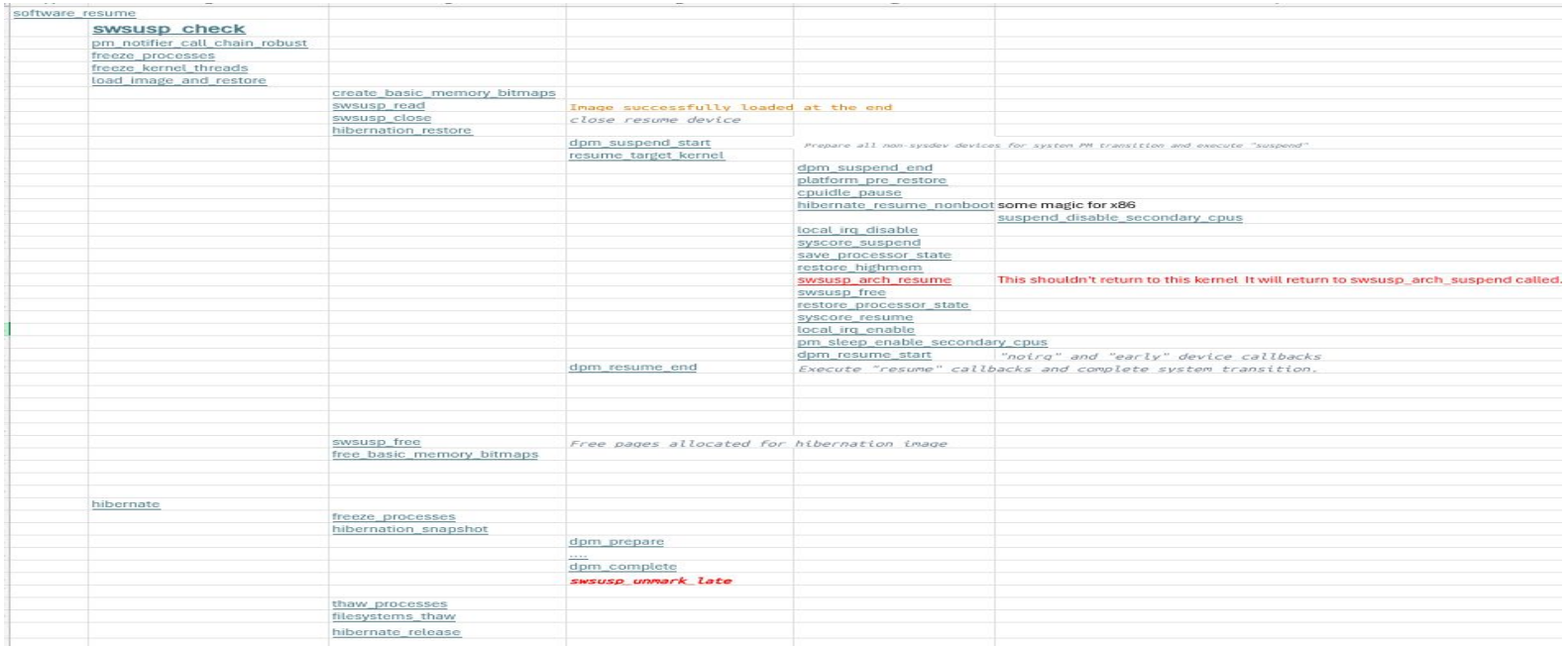
- Backup to secondary swap
- Add additional kernel command line argument
  - resume_backup_offset
- Copy the compressed image at this offset
  - Hibernate signature is preserved
    - Clear from user space (mkswap)

BACKUP SLIDES

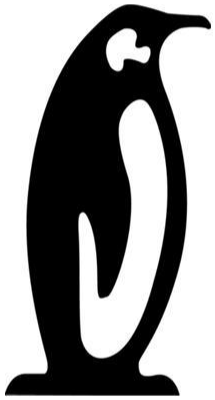
Hibernate suspend flow

hibernate				
	freeze_processes			
	hibernation_snapshot			
		dpm_prepare		
		dpm_suspend		
		create_image		
			dpm_suspend_end	
			pm_sleep_disable_secondary_cpus	
			local_irq_disable	
			syscore_suspend	
			save_processor_state	
			swsusp_arch_suspend	
			swsusp_save	
			syscore_resume	
			local_irq_enable	
			pm_sleep_enable_secondary_cpus	
			dpm_resume_start	
		dpm_resume		
		dpm_complete		
	swsusp_write	*compress and write image*		
	swsusp_free			
	power_down			
		hibernation_platform_enter		
			pm_sleep_disable_secondary_cpus	

Unmark swap from resumed kernel



printk: Allow printk_trigger_flush() to flush all types
printk: Avoid scheduling irq_work on suspend
printk: Avoid irq_work for printk_deferred() on suspend



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